

Linguistic Analysis for Conceptual Class Identification

Motivation

Linguistic analysis is a valuable source for identifying conceptual classes in a domain. Fully dressed use cases, such as the **Process Sale** use case, provide rich textual content for noun phrase mining.

Candidate conceptual classes may be found in:

- Use cases (e.g., **Sale**, **Cashier**, **Customer**)
- Other documentation
- Domain experts' mental models

Some noun phrases are attributes or concepts ignored in the current iteration (e.g., **Accounting**, **Commissions**).

Note: Natural language is imprecise; different phrases may represent the same class or attribute. This technique is recommended in combination with a **Conceptual Class Category List**.

Process Sale Use Case: Main Success Scenario

Basic Flow

1. Customer arrives at POS checkout with goods/services.
2. Cashier starts a new sale.
3. Cashier enters item identifier.
4. System records sale line item, presents item description, price, and running total. Cashier repeats steps 2-3 until done.
5. System presents total with taxes calculated.
6. Cashier informs customer of total and requests payment.
7. Customer pays; system processes payment.
8. System logs completed sale and sends sale/payment info to external **Accounting** and **Inventory** systems.
9. System prints/presents receipt.
10. Customer leaves with goods and receipt.

Alternative Flows / Exceptions

Example: Paying by Cash

- 7a.1 Cashier enters cash amount tendered.
- 7a.2 System presents balance due and opens cash drawer.
- 7a.3 Cashier deposits cash tendered and returns balance to customer.
- 7a.4 System records cash payment.

Example: Finding and Drawing Conceptual Classes

POS Domain Model (Iteration-1)

From category list and noun phrase analysis, candidate conceptual classes are identified. Focus here is on the **basic cash-only Process Sale scenario**.

Candidate Conceptual Classes:

- Sale, CashPayment, SalesLineItem, Item
- Register, Ledger, Cashier, Customer, Store
- ProductDescription, ProductCatalog, Payment, LineItem

Guideline: Associations and attributes are added later, after identifying candidate classes.

POS Domain Model Diagram

Initial UML Conceptual Classes:

- Sale → associated with SalesLineItem, Payment, Customer, Register
- SalesLineItem → linked to Item, ProductDescription, ProductCatalog
- CashPayment → linked to Sale and Cash
- Ledger → receives entries from Sale and Payment
- Cashier → performs Sale, interacts with Customer and Register

Note: This is an initial POS domain model for iteration-1. Adding attributes, multiplicities, and detailed associations comes in subsequent modeling steps.